Rank intervals from one string marking

Human Compression project

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Abstract

We apply the common contraction theorem to the principal results of Brooksbank and Leemans, *Rank reduction of string C-group representations*. The same proof gives the stated rank intervals for symmetric groups and for the retained classical-group constructions. The highest-rank classical markings remain attributed inputs. We also recover the source's finite matrix example and explain directly why contraction of its three alternating-group diagrams loses generation. No solution of the final rank-spectrum conjecture is asserted.

1 The source theorem and its iteration

The source is the complete six-page author edition arXiv:1812.01055v2, 3 May 2019, denoted by [BL]. Our common paper *Contraction of string markings through a central factor*, Theorem 3, proves its Theorem 1.1: for an irreducible rank-n string C-marking, $n \geq 4$, the condition

$$r_0 \in \langle r_0 r_2, r_3 \rangle$$

ensures that $(r_1, r_0 r_2, r_3, \dots, r_{n-1})$ is a C-marking of the same group in rank $n-1$. The common proof retains the standard end criterion and proves the central-twist and dihedral steps in full. It does not import [BL, Theorem 1.1] as an unproved premise.

If $r_2 r_3$ has finite odd order m, then $((r_0 r_2) r_3)^m = r_0$. This recovers [BL, Corollary 1.2]. For iteration use the following precise form of common Corollary 5: if $p_j = |r_{j-1} r_j|$ and $p_3, \dots, p_{t+2}$ are odd, where $0 \leq t \leq n-3$, all ranks from $n-t$ through n occur. The first test is p_3, the next p_4, and so on. Irreducibility is preserved, so each successive application has its full hypotheses.

The displayed maximum in the literal [BL, Corollary 1.3] additionally tests p_2 and is undefined if p_2 is even. The version just stated uses exactly the bonds required by its proof, permits zero steps without any parity assumption, and implies every instance where that printed maximum is defined. This indexing qualification does not change any rank interval deduced below.

2 Symmetric and alternating groups

Proposition 1. *For every $m \geq 4$, the group S_m has string C-markings of every rank $3 \leq n \leq m-1$.*

Proof. Start with the adjacent transpositions $(1\ 2), (2\ 3), \dots, (m-1\ m)$. A selected set of these transpositions generates the full symmetric group on each connected interval of the selected path, independently on the different intervals. The intersections of two such interval partitions are exactly the components of the intersection of their selected edge sets: two vertices lie in the same component precisely when every intervening edge is selected. Thus the marking has the intersection property. Its adjacent products all have order three and it generates S_m. Common Corollary 5 gives every stated rank; $m = 4$ is already the rank-three seed. □

This is the existence conclusion already proved by Fernandes and Leemans in their 2011 symmetric-group paper, and explicitly deduced in [BL, §3]. We make no claim here about uniqueness or enumeration of its highest-rank representations.

For the alternating-group remarks in [BL, §3], the necessary input is more specific than the existence of some marking. We retain the highest-rank markings of Fernandes–Leemans [FL], in the already acquired author edition arXiv:1810.12450. For $m \equiv 1 \pmod 4$, $m \geq 13$, their Theorem 4.6 supplies rank $R = (m-1)/2$ and type $[10, 3^{R-2}]$. Every bond tested by contraction is three, so common Corollary 5 yields every rank 3 through R. For even $m \geq 14$, the highest-rank instance underlying their Theorem 4.1 has rank $R = (m-2)/2$ and type

$$[5, 6, 3^{R-6}, 6, 6, 3].$$

The consecutive odd bonds beginning at the third position give $R - 6$ contractions, hence every rank from six to R. This recovers the rank-existence observation made in [BL] about that theorem. The full alternating-group classification in other congruence classes, its exact lower-rank diagrams, and the nonexistence of particular ranks do not follow from this one seed; their original constructions and classifications remain outside this deduction.

3 Classical groups: the construction inputs are retained

We first specify the complete input used for characteristic two. Brooksbank–Ferrara–Leemans [BFL, Corollary 1.5 and §5] construct, for $q = 2^k$ with $k \geq 2$ and $m \geq 2$, string C-markings of $O^+(2m, q)$ and $O^-(2m, q)$ in rank $2m$, and of $\mathrm{Sp}(2m, q)$ in rank $2m + 1$. The adjacent products in these markings all have order $q + 1$.

This input includes generation, the intersection property, and the choice of Witt type. In the exact primary edition arXiv:1709.02219v2, 10 March 2018, the coefficients in §5 are chosen so that the two-dimensional products have order $q + 1$; Lemma 5.3 and Proposition 3.3 supply generation and intersections, and the proof of Corollary 1.5 chooses each Witt type and a nonsingular radical in odd dimension. None of those construction arguments is reclassified as a consequence of contraction.

Theorem 2. *For integers $k, m \geq 2$, $\mathrm{Sp}(2m, 2^k)$ has string C-markings in every rank $3 \leq n \leq 2m + 1$, and each $O^{\pm}(2m, 2^k)$ has such markings in every rank $3 \leq n \leq 2m$.*

Proof. Each retained highest-rank marking is irreducible: its adjacent order is $q + 1 \geq 5$. That order is odd. Apply common Corollary 5 until rank three. The group is preserved at each contraction. This proves both clauses of [BL, Theorem 1.4] with exactly the stated field and dimension restrictions. □

No assertion for $q = 2$, no transfer to the derived orthogonal subgroup, and no claim of maximality of these rank intervals is being made.

For the odd-characteristic result we retain the reflection construction of Monson–Schulte, *Reflection groups and polytopes over finite fields, III*, §3 and Theorem 3.1, together with its orthogonal-group identification, as invoked in [BL, §4]. The required form of that input is: for $p \geq 5$ prime and $d \geq 3$, reduction of the crystallographic string $[3, \infty^{d-2}]$ gives a rank-d C-marking of an index-two subgroup $M \leq O(d, p)$, with adjacent orders $[3, p^{d-2}]$. The original Monson–Schulte PDF is not an acquired local primary in this packet; its construction and group-identification theorem are explicitly retained through the source's statement. This is a proof dependency, not newly certified computation.

There is no hidden degeneracy in this chosen family. The integral Gram matrix has diagonal entries 2, first adjacent entries -1, remaining adjacent entries -2, and zero elsewhere. Its leading determinants satisfy $D_2 = 3$, $D_3 = -2$, $D_4 = -16$ and

$$D_j = 2D_{j-1} - 4D_{j-2} \quad (j \geq 3), \qquad D_{j+3} = -8D_j \quad (j \geq 2).$$

The second identity follows from the first or by induction from the three initial values. Every D_d has no prime divisor other than two and three. Consequently the form is nondegenerate after reduction at every $p \geq 5$. This calculation verifies the form condition; it does not replace the retained generation or intersection theorem.

Theorem 3. *For every $d \geq 3$ and prime $p \geq 5$, an index-two subgroup $M \leq O(d, p)$ has string C-markings in every rank $3 \leq n \leq d$.*

Proof. All adjacent orders of the retained rank-d marking are odd and greater than two. Common Corollary 5 preserves its group M and provides every rank down to three. For $d = 3$ the seed itself is the required marking. This is [BL, Theorem 4.1]. The orthogonal type and the index-two subgroup are those of the retained reflection construction; no identification with an arbitrary prescribed simple projective group is inferred. □

4 The matrix example outside the recovery hypothesis

The membership condition in common Theorem 3 is sufficient for generation, but it is not necessary for every successful contraction. We retain the literal example of [BL, end of §2] and verify it with the complete program `check_examples.g`. Over $\mathbb{F}_3$ put

$$F = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{pmatrix},$$

and use row-vector action with the following matrices:

$$r_0 = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, r_1 = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix},$$

$$r_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 2 & 1 \end{pmatrix}, r_3 = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives $r_i F r_i^{\mathsf{T}} = F$. There are exactly 21 singular vectors, including zero, so this nonsingular four-dimensional form has minus type. The generated matrix group has order 1440, equal to $|O^-(4, 3)| = 2 \cdot 3^2(3^2 + 1)(3^2 - 1)$; hence it is the full isometry group, conventionally identified with $\mathrm{SL}(2, 9) : 2$. The last identification and the orthogonal order formula are standard classical-group facts, not needed for the intersection check.

The program verifies all 136 unordered pairs of parabolics of the four-mark tuple and all 36 for $(r_1, r_0 r_2, r_3)$, including identical subgroups and the trivial subgroup. It checks the types $[4, 4, 6]$ and $[6, 6]$, equality of the two generated groups, and $r_0 \notin \langle r_0 r_2, r_3 \rangle$. This is a complete finite verification of the displayed example, using trusted GAP group computation; it is not an argument from samples for common Theorem 3.

5 The three A_{11} diagrams and the remaining question

Number the eleven vertices of each path in [BL, Figure 1] from left to right. Its ten successive edges have the following label sets; a pair in one entry denotes the displayed double edge.

	1	2	3	4	5	6	7	8	9	10
I	$\{0,2\}$	1	0	1	2	3	4	5	4	$\{3,5\}$
II	0	1	$\{0,2\}$	1	2	3	4	5	4	$\{3,5\}$
III	0	1	$\{0,2\}$	1	2	3	4	$\{3,5\}$	4	5

The involution r_i is the product of the path transpositions carrying label i. Our complete program checks that each tuple generates A_{11}, that all 2080 unordered parabolic pairs satisfy IP, and that the three types are $[5,3,6,3,5]$, $[5,5,6,3,5]$, and $[5,5,6,5,5]$. Their different types, even after reversal, distinguish the three markings. The statement that these exhaust the rank-six cases, and that A_{11} has rank spectrum $\{3,6\}$, remains the prior classification cited in [BL, §3], Fernandes–Leemans–Mixer [FLM]. The finite check does not prove exhaustion or the absence of ranks four and five.

For left contraction, the common transposition carrying labels 0 and 2 cancels in r_0r_2. It is the first path edge in row I and the third in rows II and III. No other generator crosses that edge, so the new subgroup is intransitive, with orbit sizes $(1,10)$ or $(3,8)$. From the right, the analogous common edge of labels 3 and 5 cancels at the tenth edge in rows I and II and at the eighth in row III. The corresponding orbit sizes are $(10,1)$ or $(8,3)$. Thus neither direction generates A_{11}. This proves the specific limitation in [BL] directly from all three literal diagrams.

Conjecture 5.1 of [BL] asks whether A_{11} is the only finite simple group whose C-marking rank spectrum has a gap. Neither the contraction theorem nor these examples settle that question. We retain the distinction between an obstruction for a specified marking and an existence statement over all markings of a group.

6 Proof and computational boundaries

The complete common theorem, its parity calculation and the applications above recover [BL, Theorem 1.1, Corollaries 1.2–1.3, Theorems 1.4 and 4.1] under the explicitly retained highest-rank construction inputs. The iteration statement uses the precise indexing explained in Section 1. The symmetric and selected alternating consequences and both families of finite examples are also included. We do not claim to reprove the retained classical seeds, the alternating classification, or the rank-gap conjecture.

The one whole GAP program is required for the two finite-example certificates. The uniform proofs do not depend on its output. The four declarations in `RankContraction.lean` prove the odd-power recovery identity, membership of the recovered generator, recovery of its other factor, and exclusion from a generated subgroup by a homomorphism. They are checked with Lean 4.32.2 and the recorded Mathlib revision. They do not formalize the standard end criterion, the induction proving the C-property, the classical-group seeds or the complete source-level application. No compression factor or historical-priority claim is attached to this draft.

References

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